p-adic Analysis 4

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Problem 1

The first digit is 2 since $2^2 = 4 \equiv -3 \pmod 7$. Lifting to higher powers of 7 gives us

$$\sqrt{-3} \in \mathbb{Q}_7 = \ldots 1256052.$$

Problem 2

Assume $p \nmid n$ and $\alpha \equiv 1 \pmod p$. Consider

$$f(x) = x^n - \alpha.$$

Since $\alpha \equiv 1 \pmod p$,

$$f(1) = 1 - \alpha \equiv 0 \pmod p.$$

Also,

$$f'(x) = nx^{n-1},$$

so $f'(1) = n$. Since $p \nmid n$,

$$f'(1) \not\equiv 0 \pmod p.$$

Therefore, by Hensel's lemma, there exists $x \in \mathbb{Z}_p$ such that

$$x^n = \alpha.$$

Problem 3

a must be even. Using Hensel's lemma on

$$f(x) = x^2 + x + a,$$

we get

$$f(x) \equiv x^2 + x + a \equiv 2x + a \equiv a.$$

If a is odd, then $f(x) \equiv 1 \pmod 2$ for both $x = 0, 1$. There is no root $\pmod 2$, so no root in $\mathbb{Z}_2$ (and therefore $\mathbb{Q}_2$ by rational root theorem). If a is even, $f(x) \equiv 0 \pmod 2$ for both $x = 0$ and $x = 1$. Also,

$$f'(x) = 2x + 1 \equiv 1 \pmod 2,$$

so Hensel's lemma applies, and $x^2 + x + a$ has a root in $\mathbb{Z}_2$.

Problem 4

Trying each $x \in \{0, 1, 2, 3, 4, 5, 6\}$, we have that $x \equiv 2, 3, 4, 5 \pmod 7$ satisfy $f(x) \equiv 0 \pmod 7$. Also,

$$f'(x) = 4x^3 + 2x,$$

which is nonzero $\pmod 7$ for all $x = 2, 3, 4, 5$. By Hensel's lemma, each gives a root of $f(x)$, so there are 4 roots.

Problem 5

In $\mathbb{R}$, $x = \sqrt{2}$ is a root. But since $\sqrt{2}, \sqrt{17}, \sqrt{34}$ are all irrational, there are no roots in $\mathbb{Q}$.

First, assume $p \neq 2, 17$. Then

$$\frac{34}{p} = \frac{2}{p} \cdot \frac{17}{p} \equiv (-1)^2 = 1.$$

So 34 is a square $\pmod p$, and by Hensel's lemma, 34 is a square in $\mathbb{Q}_p$.

Now, assume $p = 2$. Since $17 \equiv 1 \pmod 2$, 17 is a square in $\mathbb{Q}_2$.

Finally, assume $p = 17$. 2 is a square, since $6^2 \equiv 36 \equiv 2 \pmod{17}$. The derivative of $x^2 - 2$ at 6 is 12, so by Hensel's lemma, 6 lifts to a root of $x^2 - 2$ in $\mathbb{Q}_{17}$.

Problem 6

For $\mathbb{R}$: take $x = 3$ and $y = \pm 4\sqrt{2}$.

For $\mathbb{Q}_p$: let $p = 2$. Since $17 \equiv 1 \pmod{16}$, 17 has a fourth root in $\mathbb{Q}_2$. So choose x with $x^4 \equiv 17$ and take $y = 0$.

Now, let $p = 17$. We can take $x = 1$ and $y = 3$, so that

$$x^4 - 17 \equiv 1, \quad 2y^2 \equiv 1 \pmod{17}.$$

Additionally,

$$\frac{\partial}{\partial y}(x^4 - 17 - 2y^2) = -4y,$$

which is $-12 \not\equiv 0 \pmod{17}$ at $y = 3$, so by Hensel's lemma, this lifts to a solution in $\mathbb{Q}_{17}$.

I couldn't figure out the $p \neq 2, 17$ case.

Problem 7

Assume

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad a_n \in \mathbb{Z}_p.$$

Also assume α_0 satisfies

$$f(\alpha_0) \equiv 0 \pmod{p}, \quad f'(\alpha_0) \not\equiv 0 \pmod{p}.$$

Pick any lift x_1 of α_0 and assume x_n is known. We wish to find

$$x_{n+1} = x_n + p^n t, \quad t \in [p].$$

Using the taylor expansion,

$$f(x_n + p^n t) = f(x_n) + p^n t f'(x_n) + p^{2n}(\cdots) + \cdots .$$

The higher terms vanish mod p^{n+1}, so

$$f(x_{n+1}) \equiv f(x_n) + p^n t f'(x_n) \pmod{p^{n+1}}.$$

Since $f(x_n) \equiv 0 \pmod{p^n}$, we have that $f(x_n) = p^n c$ for some c. So we need

$$p^n(c + t f'(x_n)) \equiv 0 \pmod{p^{n+1}},$$

$$c + t f'(x_n) \equiv 0 \pmod{p},$$

which is solvable for t, since $f'(x_n) \equiv f'(\alpha_0) \not\equiv 0 \pmod{p}$. Continuing to infinity, we get $x \in \mathbb{Z}_p$ where $f(x) = 0$.